

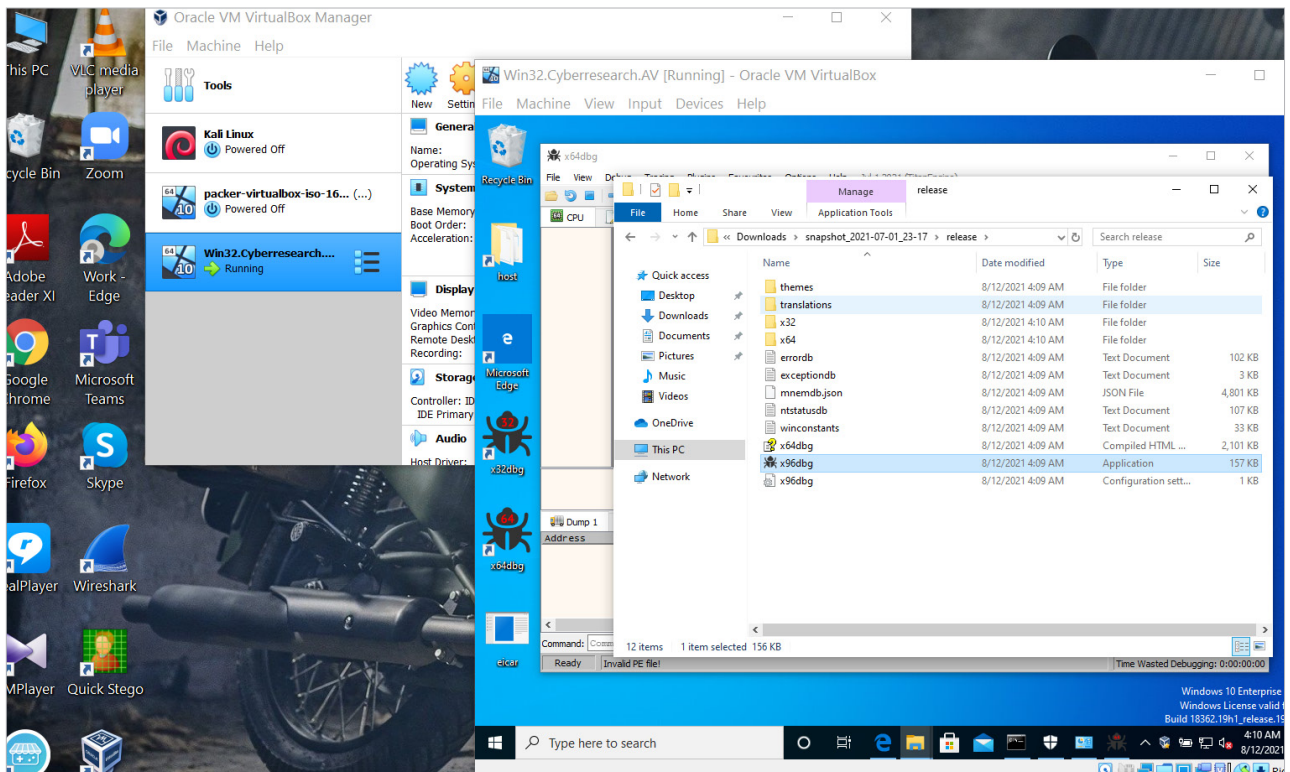


Figure 7: Drag and drop the executable file into the debugger

Packer-created images, embracing modern configuration management. The Packer images allow machines that are fully loaded and configured to start in seconds. They can run production in AWS, staging/QA in a private cloud like OpenStack, and development in desktop virtualisation solutions like VMware or VirtualBox because they build similar images for many platforms. Add the downloaded *exe* file to environment variables in the *Advanced System Settings*, while installing Packer 1.2.3.

**Install Malboxes and setup tools through CLI:** Setuptools is a set of improvements to the Python *distutils*. It makes creating and developing Python easier for developers, particularly those that rely on other packages. Since Malboxes have dependencies, we can use setuptools to install them correctly.

Malboxes builds malware analysis on Windows virtual machines, providing an environment to test malware. Figure 3 shows the commands that have to be

entered into the command prompt.

Next, check for Malboxes installation and create a box. As shown in Figure 4, this will build a Windows 10, 64-bit VM. We can check for other VMs by entering 'malboxes list' in the CLI.

Now create a Vagrant file for malware analysis by typing in 'vagrant up'.

This command creates and configures guest machines according to your Vagrant file. This is the single most important command in Vagrant (can be viewed in Figure 5), since this is how any Vagrant machine is created.

In the next step, spin up the new VM using the above syntax; for example: 'malboxes spin win7\_32\_analyst newWin32VM'.

The ISO file can be found in the virtual box by double clicking on the new ISO file, or just running it by clicking on *run*. The VM should now start booting. Make sure sufficient RAM and system resources are

provided to the VM.

## Sample malware analysis

Let us now analyse a sample malware executable file in the built-in WinDBG on the Windows virtual machine. The steps to carry out the analysis are as follows.

**Download a sample malware file:** There are multiple resources from which we can download suitable malware executables for analysis, one of which is the GitHub repository called 'Malware-Samples by fabricag72'. These samples were caught by the owner of the repository using various honeypots. TekDefense has a bunch of executable malware samples as well, as seen in Figure 6.

**Disable Defender and prepare for analysis:** Find 'disable\_defender' under the tools and execute it. This will disable the Defender on the VM.

Open a debugger such as WinDBG that comes with the virtual machine. It should be titled 'x64dbg'.